A Complex Adaptive Systems Network Model of Canada's Housing Affordability and Rent Crisis

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Abstract — Canada's housing affordability crisis exhibits properties inconsistent with equilibrium theory: feedback loops, non-linear tipping points, and persistent emergent spatial inequality. This paper presents a Complex Adaptive Systems (CAS) network/graph model of Canada's housing affordability and rent crisis, calibrated using open-source data from Statistics Canada and the Canada Mortgage and Housing Corporation (CMHC) Rental Market Survey (2023). The model represents 150 household agents, 6 neighbourhood nodes (one per major Census Metropolitan Area), and 20 employer nodes as an interconnected weighted graph, with edges encoding tenancy and employment relationships. Six experiments, three demonstrating emergence and three demonstrating self-organization are conducted by systematically varying four core parameters: rent increase rate, income growth rate, affordability threshold, and newcomer employment bias. Results show that macro-level displacement inequality and affordability collapse emerge spontaneously from local agent interactions. Displacement rates scale from 11.33% to 100% as rent increase rates rise from 2% to 16%, and a newcomer displacement gap of up to 36 percentage points emerges from purely local hiring bias. Early rent cap policy interventions reduce final displacement by 75% relative to no intervention, demonstrating strong path dependency consistent with CAS resilience theory.

I. Introduction

Over the last decade, Canada's rental housing sector has seen significant structural changes. National average rental costs for a one-bedroom apartment reached $1,519, but real income growth remained below 3% per year [2]. In cities like Toronto and Vancouver, vacancy rates plummeted to 1.5% and 0.9%, respectively, resulting in considerable rental market pressure [1]. Simultaneously, newcomers to Canada confront compounded disadvantages: credential non-recognition, hiring discrimination, and precarious work result in salaries much lower than the national median, leaving them more vulnerable to housing instability [3].

Conventional economic models view housing markets as systems that move toward equilibrium via price signals. However, the Canadian housing crisis exhibits traits that contradict equilibrium theory, including feedback loops, non-linear tipping points, and emergent spatial inequity that persists even when individual agents act rationally. According to research, network structure can enhance inequality through segregation dynamics in the absence of an explicit discriminatory mechanism [4]. These are characteristics of Complex Adaptive Systems (CAS), in which macro-level patterns emerge from local interactions between various agents that lack central coordination [6].

This paper models Canada's housing affordability crisis as a CAS using a network/graph approach implemented in Python 3.11.14. The model is calibrated using open-source Canadian datasets and used to conduct six experiments demonstrating emergence and self-organization. Section II examines related work and in Section III, it describes the model. Section IV shows the results while Sections V and VI discuss findings and conclude.

II. Related Works

Walks [1] examines financial landlords in Toronto's rental market, finding that financialized ownership drives rents above market rates particularly in lower-income neighbourhoods. This caught my attention due to the fact that in order to make revenue gains, a "value add" strategy was adopted. This motivates the model's vacancy-weighted rent increase mechanism.

Ferreira [2] conducted an interview and wrote an article about Canada's housing affordability crisis, discovering that average rental prices for a one-bedroom apartment have reached $1,519 nationally and that both the government and banking interests benefit from the current housing shortage while affordability deteriorates. The figure serves as a baseline against which comparable apartments in other provinces can compare. This provides the Canadian empirical context motivating the model's rent calibration and policy shock experiment named as S3.

The findings and data revealed a persisting difference between recent immigrants and Canadian-born workers. Immigrants did poorly in the labor market (E. Crossman, F. Hou, and G. Picot, [3]). This directly influences the newcomer_employment_bias parameter and the inverse-wage employer edge assignment in the model.

Gergő [4] uses real urban network data to show that income disparity increases sharply when social network segregation interacts with urban geometry, indicating that network structure alone can magnify inequality. This conclusion has direct implications for the model's network building and the emergence of newcomer displacement gaps in E2.

In Schelling-type models, Will and Hiroki [5] demonstrate how network topology greatly influences the establishment of segregation, discovering that scale-free topologies speed up clustering. This provides information on the network construction's weighted CMA population distribution.

Thakkar [6] characterizes the affordable housing scarcity as a CAS issue, saying that agents such as politicians must account for feedback loops and path dependency. The rent cap

policy shock experiment (S3) and the route dependency conclusions in Section V are directly motivated by this.

III. Methodology

A. Network Model Design

The model represents Canada's urban housing market as a weighted undirected graph G = (V, E), implemented with the NetworkX library in Python 3.11.14. The node set V contains three agent types. Household nodes (150 agents) carry annual income drawn from CMA-specific distributions centred on Statistics Canada 2021 Census medians, newcomer status (assigned with 30% probability), rent burden (monthly rent / monthly income), and housed status. Neighbourhood nodes (6 agents, one per CMA) carry average rent initialized from CMHC 2023 data, vacancy rate from CMHC 2023, and affordability score. Employer nodes (20 agents) carry a wage level and a newcomer hiring bias parameter.

The edge set E encodes two relationship types: household-to-neighbourhood tenancy edges, and household-to-employer employment edges. CMAs modelled are Toronto ($2,100/mo baseline rent, 1.5% vacancy), Vancouver ($2,300, 0.9%), Calgary ($1,700, 2.1%), Ottawa ($1,650, 2.2%), Montreal ($1,350, 3.0%), and Winnipeg ($1,200, 2.8%), calibrated directly from CMHC 2023 rental survey data.

B. Core Parameters

The four parameters that govern all experiments are as follows: (P1) rent_increase_rate—annual fractional rent increase, vacancy-weighted to reflect supply pressure; (P2) income_growth_rate—annual fractional household income growth, reduced for newcomers by factor (1 − bias × 0.4); (P3) affordability_threshold—maximum acceptable rent-to-income ratio, set to the CMHC core housing need definition of 30%; and (P4) newcomer_employment_bias—fractional hiring disadvantage for newcomer agents, calibrated from Statistics Canada immigrant earnings data. P1=0.08, P2=0.03, P3=0.30, and P4=0.35 are the baseline values. The parameters were limited to 4 to avoid ambiguous results and time consumption.

C. Step Function and Datasets

Each time step represents one year. The update sequence is: (1) neighbourhood rents increase at rate P1, amplified when vacancy falls below 3%; (2) household incomes grow at rate P2; (3) rent burdens are computed pre-displacement; (4) households whose burden exceeds P3 attempt relocation — self-organization emerges here as agents independently scan the network for affordable vacancies; (5) households that cannot relocate are displaced; (6) neighbourhood affordability scores are updated. The following datasets were utilized, and despite encountering a merged header issue, the model worked once it was resolved: Statistics Canada, Table 46-10-0072-01 (housing acceptability by immigrant status), Statistics Canada Table 18-10-0205-01 (Housing Price Index by CMA), and CMHC Rental Market Survey data tables (average rents and vacancy rates, 2023).

IV. Experiments

All experiments run for 10 time steps with 150 households, 6 neighbourhood nodes, and 20 employers (seed = 42 for reproducibility).

A. E1 — Rent Rate vs. Displacement Emergence

P1 was altered at five different levels: 2%, 5%, 8% (baseline), 12%, and 16%. The final displacement rates were 11.33%, 29.33%, 52.0%, 86.67%, and 100%, respectively (Figure 1). Displacement is not pre-programmed; it results from local rent burden calculations and relocation decisions. At each stage, neighbourhood rents rise while household income climbs slowly. This requires household agents to adjust their rent burden over time. If the burden exceeds 30%, households attempt to relocate by looking for cheaper available neighbourhoods. However, because vacancies are few and rents are rising, many people are unable to find alternatives and are displaced, leading displacement to escalate rapidly as more households reach the threshold together. Toronto reached $4,648 per month under the baseline, demonstrating emergent spatial rent inequality. The gap is not caused by any central law but from the outcome of local rent revisions linked to vacancy rates and the uneven distribution of households across neighbourhoods.

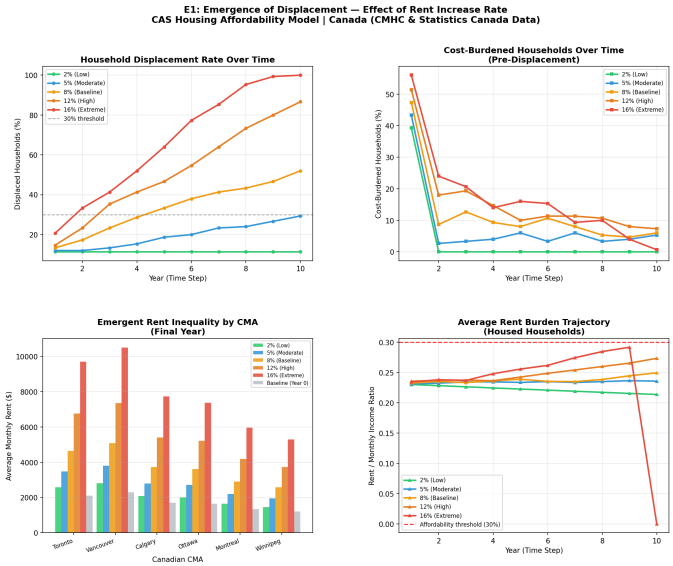

Fig. 1. Rent Rate vs. Displacement Emergence

B. E2 — Newcomer Bias vs. Income Gap Emergence

P4 was varied from 0.0 to 0.75. At zero bias the newcomer displacement gap was −2.9 percentage points. As bias increased the difference widened non-linearly: 3.3pp at bias = 0.15, 8.0pp at baseline (0.35), 31.3pp at 0.55, and 36.0pp at 0.75 (Fig. 2). A macro-level newcomer inequality gap emerges entirely from local hiring decisions and no explicit discrimination mechanism exists in the model.

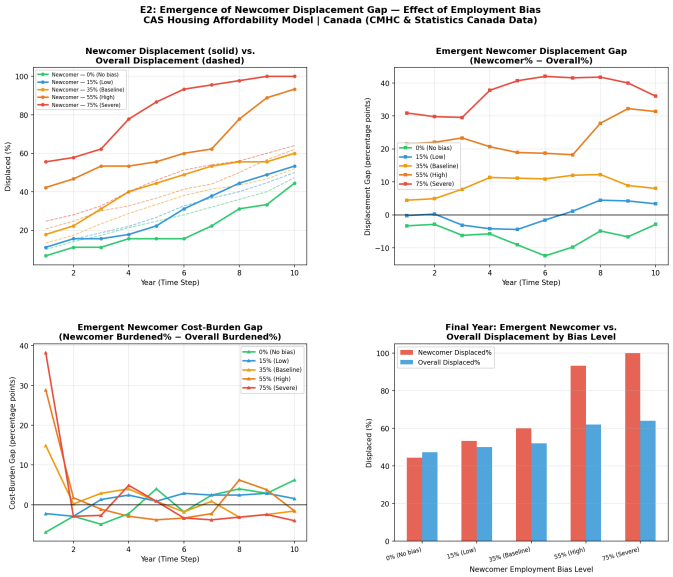

Fig. 2. Newcomer Bias vs. Income Gap Emergence

C. E3 — Rent-Income Gap vs. Affordability Collapse

P1 and P2 were co-varied: balanced (3%/3%), mild (5%/3%), moderate baseline (8%/3%), severe (12%/2%), and crisis (16%/1%). Displacement: 12%, 29.33%, 52%, 94.67%, and 100%. The crisis scenario resulted in a total affordability collapse (Fig. 3). By year 6, affordability scores in Toronto and Vancouver had fallen below 0.30, indicating the emergence of spatial collapse zones.

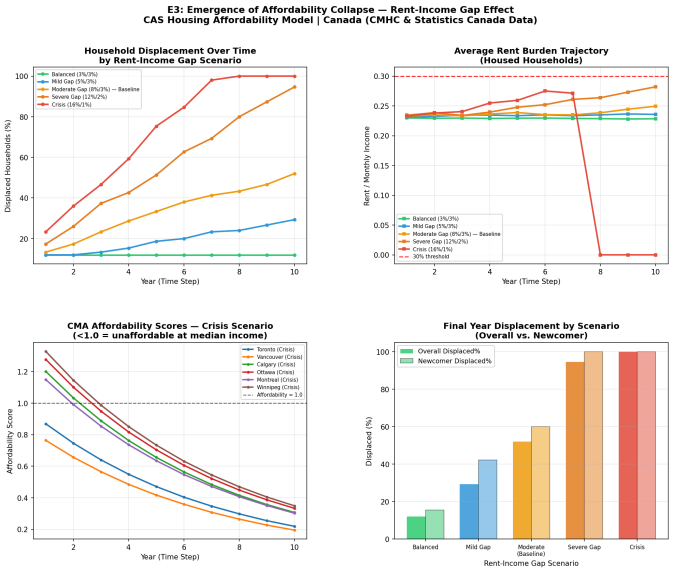

Fig. 3. Rent-Income Gap vs. Affordability Collapse

D. S1 — Threshold vs. Household Self-Sorting

P3 varied from 20% to 40%. A self-organization concentration index peaked at threshold = 0.30 (concentration = 1.689), exactly the CMHC core housing need definition (Fig. 4). The CMHC 30% threshold is the natural self-organization tipping point where households most actively self-sort across CMAs.

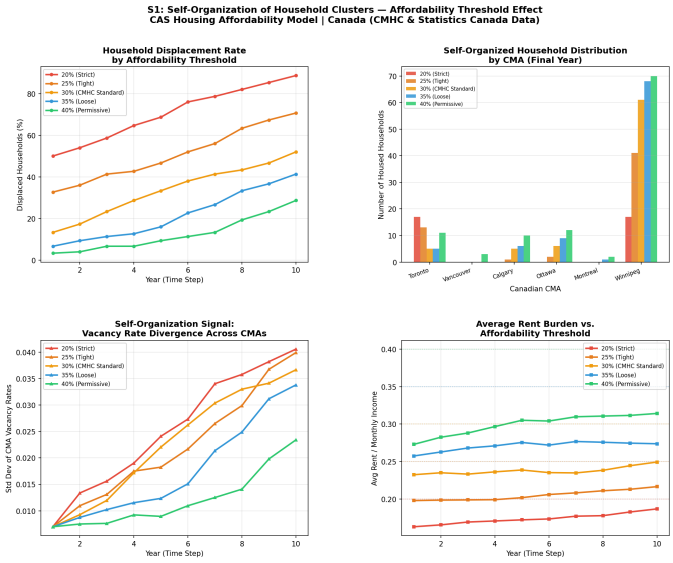

Fig. 4. Affordability Threshold vs. Household Self-Sorting

E. S2 — Employer Removal vs. Labour Network Reorganization

Highest-wage employer nodes were progressively removed (0% to 70%). Newcomer displacement grew from 60.0% to 75.56%, while the income gap narrowed from $22,008 to ~$19,000 as all incomes fell together (Fig. 5). Newcomers clustered in low-wage peripheral nodes — self-organized labour market stratification driven purely by the network's edge-assignment rules.

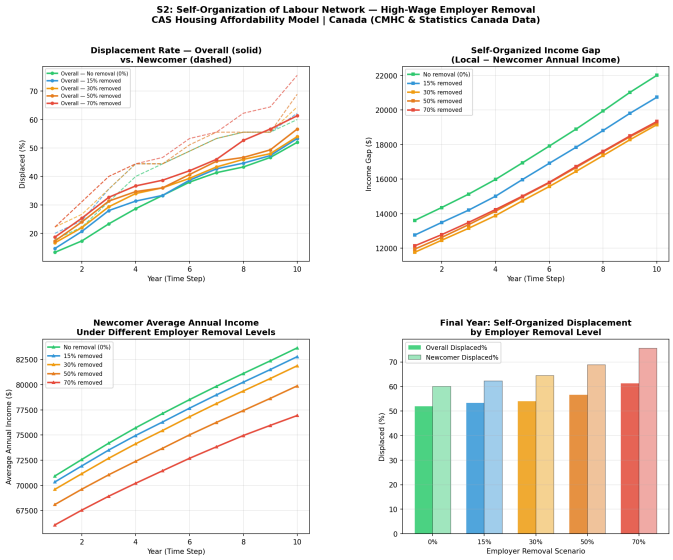

Fig. 5. Employer Removal vs. Labour Network Reorganization

F. S3 — Rent Cap Shock vs. Network Re-equilibration

A rent cap (P1 = 0%) was applied at years 2, 4, 6, and 9. Final displacement: no cap 52%, year-2 cap 13.33%, year-4 23.33%, year-6 33.33%, year-9 43.33% (Fig. 6). Average rent stabilized at $1,857 under early cap versus $3,764 with no cap, demonstrating strong path dependency and displacement locked in early cannot be reversed by late intervention.

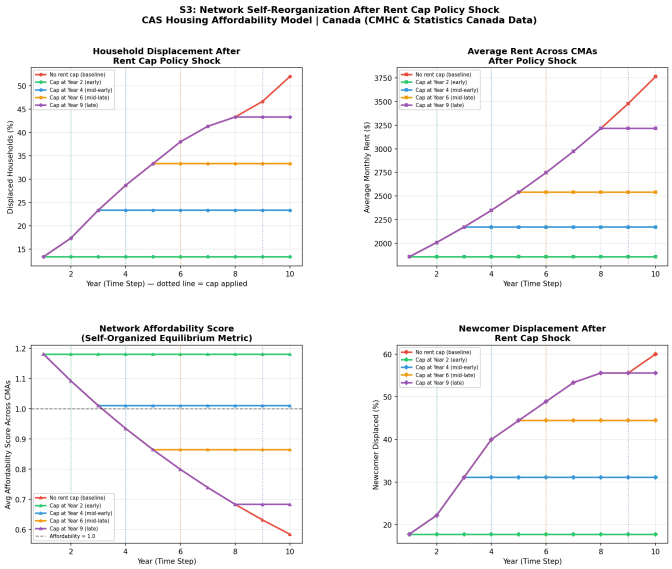

Fig. 6. Rent Cap Shock vs. Network Re-equilibration

V. Discussion

The six experiments demonstrate that Canada's housing affordability crisis exhibits both defining CAS properties.

On emergence: E3 reveals non-linear threshold dynamics, modest rent growth (5%) produces 29% displacement while doubling rent growth to 10% produces 52%, a disproportionate escalation consistent with CAS tipping point behaviour. E2 shows that newcomer inequality emerges from purely local hiring decisions: even a subtle bias of 0.15 generates a 3.3pp displacement gap, a macro pattern with no macro cause. In a lower rent city like Winnipeg, slower rent growth and slightly higher vacancy mean fewer households hit the affordability threshold at once, so displacement rises more gradually.

In high-pressure markets like Toronto, low vacancy and higher baseline rents push many households past the threshold simultaneously, triggering sharp, non-linear spikes in displacement. Similarly, even small hiring biases can widen inequality faster in cities like Toronto, where tighter conditions amplify the effects of local disadvantages.

On self-organization, S1's concentration index identifies the CMHC 30% threshold as a natural self-organization tipping point, with the system reorganizing most aggressively at this precise limit. S3's path dependency finding is the most policy-relevant result: a rent cap in year 2 results in 13.33% final displacement, but the same cap in year 9 results in 43.33%, a 30 percentage point change from an equivalent policy. In CAS housing markets, time is just as essential as intervention itself. On this point, the remarkable 30% point difference between the second and ninth years of S3 demonstrates how, over time, the decisions of agents such as policymakers, landlords, and so on influence decisions made by household agents.

VI. Conclusion

This paper presented a network/graph CAS model of Canada's housing affordability and rent crisis, calibrated using open-source Statistics Canada and CMHC datasets. Six experiments demonstrated that macro-level displacement inequality, affordability collapse, and labour market stratification are emergent properties of local agent interactions, and that the housing network self-organizes into distinct household clusters without central coordination. Key findings include the CMHC 30% threshold as a self-organization tipping point, non-linear emergence of newcomer displacement gaps from local hiring bias, and strong path dependency in the network's response to rent cap shocks. These findings indicate that Canada's housing crisis requires early, sustained policy intervention that accounts for the feedback loops and threshold dynamics of complex adaptive systems rather than point-in-time equilibrium corrections.

This paper has a few limitations that should be noted. First, the model relies on six discrete CMA nodes rather than a continuous geographical grid, which simplifies intracity dynamics. Second, income and rent distributions are initialized using aggregate data rather than household-level microdata, resulting in smoothing effects. Third, the model does not consider supply-side factors such as new building, zoning, and developer activity, which are major contributors to Canada's shortage. These limitations arise because the model simplifies time and agent behavior (e.g., set rules, no adaptive policy, or landlord strategies), preventing it from fully capturing how decisions evolve and interact over extended periods. Future work should include the addition of dynamic, learning agents (policymakers, landlords, and developers) as well as the extension of the simulation with richer, time-dependent data to show how interventions and behaviors evolve over time.

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